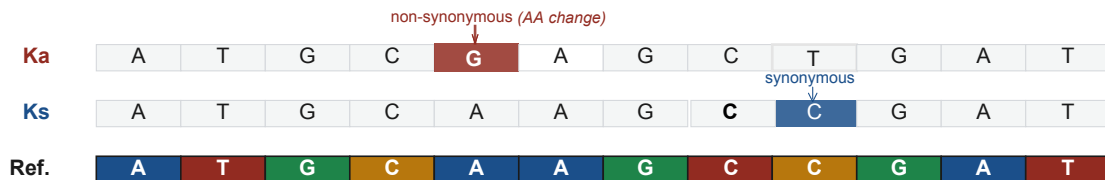


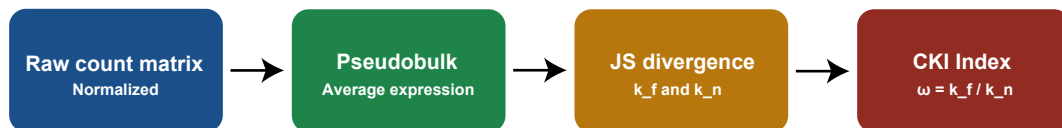
(a)



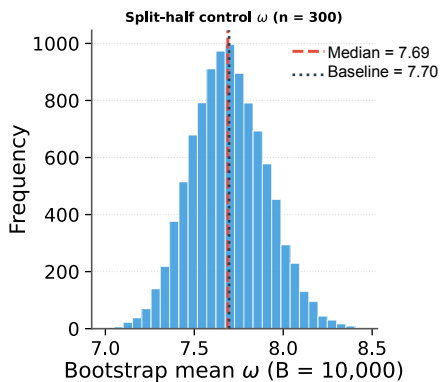
$$\omega = \frac{K_a}{K_s} \quad \omega > 1: \text{positive selection} \quad \omega \approx 1: \text{neutral drift} \quad \omega < 1: \text{purifying selection}$$

$$\text{CKI } \omega = \frac{k_f}{k_n} = \frac{\text{JS}(\text{softmax}(\epsilon_A[I]), \text{softmax}(\epsilon_B[I]))}{\text{JS}(\text{softmax}(\epsilon_A[H]), \text{softmax}(\epsilon_B[H]))} \quad \omega: \text{transcriptomic divergence relative to the baseline}$$

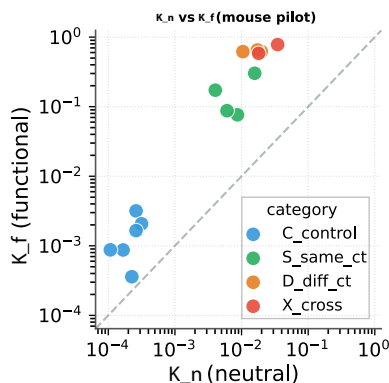
(b)



(c)



(d)



(e)

